

STAT 314500 - Applied Partial Differential Equations
Assignment 3: Term-by-term differentiation and integration of Fourier series

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapters 2 and 3 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1 Consider the heat equation describing the temperature distribution $u = u(\theta, t)$ on a thin circular wire with radius r :

$$\begin{aligned}\frac{\partial u}{\partial t} &= \frac{k}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\ u(\pi, t) &= u(-\pi, t) \\ \frac{\partial u}{\partial \theta}(\pi, t) &= \frac{\partial u}{\partial \theta}(-\pi, t) \\ u(\theta, 0) &= f(\theta).\end{aligned}\tag{1}$$

1. Compute the equilibrium temperature by solving $\frac{k}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$. Use the initial and boundary conditions stated in the original equation (1) to identify the equilibrium temperature uniquely. (3 points)
2. Verify that the solution to (1) is given by

$$u(\theta, t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\theta) e^{-\frac{n^2 kt}{r^2}} + \sum_{n=1}^{\infty} b_n \sin(n\theta) e^{-\frac{n^2 kt}{r^2}}.$$

What are the coefficients a_0, a_n and b_n ? Use this to find the equilibrium temperature distribution on the wire. (3 points)

Question 2 Solve the 1D heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = f(x)$$

subject to the following boundary conditions:

1. $u(0, t) = u(L, t) = 0$ (3 points)
2. $u(0, t) = 0, \frac{\partial u}{\partial x}(L, t) = 0$ (3 points)
3. $\frac{\partial u}{\partial x}(0, t) = 0, \frac{\partial u}{\partial x}(L, t) = 0$ (3 points)

Question 3 Consider 1D the heat equation

$$\begin{aligned}\frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2} \\ u(x, 0) &= f(x) \\ \frac{\partial u}{\partial x}(0, t) &= 0 \\ \frac{\partial u}{\partial x}(L, t) &= 0.\end{aligned}$$

Solve the heat equation above by constructing a Fourier cosine series. Assume that u and $\partial_x u$ are continuous and $\partial_{xx} u$ and $\partial_t u$ are piecewise smooth. Justify all differentiations of infinite series. (10 points)

Question 4

Consider the nonhomogeneous heat equation (with a steady heat source)

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} + g(x)$$

with initial condition

$$u(x, 0) = f(x)$$

and boundary conditions

$$u(0, t) = 0, \quad u(L, t) = 0.$$

Solve this equation in the following manner. Assume that a continuous solution with continuous derivatives exists. Expand the solution as a Fourier sine series (i.e, use the method of eigenfunction expansion). Assume that $g(x)$ can be expanded as a Fourier sine series as well and solve for the Fourier sine series of the solution. Justify all differentiations with respect to x . (10 points)

Question 5 The Fourier series of $f(x) = x, x \in [-L, L]$ is given by

$$x \sim \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}, \quad B_n = \frac{2L}{n\pi} (-1)^{n+1}.$$

Use this to derive the

1. Fourier cosine series of x^2 . (5 points)
2. Fourier sine series of x^3 . (5 points)
3. Can this be generalized to derive the Fourier sine series of x^{2m+1} ? Explain. (5 points)